

Fortress AI

System & Dashboard Whitepaper

External technical review - Neural Ops UI v2 - Not investment advice

Fortress AI - System & Dashboard Whitepaper

Document purpose: External technical review of the Fortress AI platform, its operational architecture, and every dashboard panel with metric definitions, data provenance, and trading relevance.

Version: Neural Ops UI v2 (May 2026)

Primary dashboard: Flask app at port 8050 (FORTRESS_AI_DASHBOARD_PORT)

Disclaimer: Research and paper-trading infrastructure. Not investment advice.

1. Executive summary

Fortress AI is a multi-agent quantitative trading platform built around three distinct execution paths:

Layer	Agent	Role		LLM?
Intraday skim	Skim Swarm	Always-on rule-based	1-minute long/short	No
Strategic AI	Unified AI Agent	Periodic portfolio decisions	LLM-driven	Yes (DeepSeek)
Legacy SPY	SPY Intraday Agent	Ladder-based session trading (optional; disabled when primary)	SPY skim is	No

The **Command Center dashboard** is the single operational surface for monitoring all three paths, diagnosing why trades did or did not occur, tracking P&L, governing self-improvement, and comparing Fortress AI against a parallel **Classic** rules-based stack (trading-bot).

Design principles:

- Per-symbol independence - each skim ticker owns its parameters, pattern disables, and learning state.
- Deterministic hot-path adaptation - live recursive improvement without LLM latency in the skim loop.
- Tiered self-improvement - parameter tuning (Tier 0-1), prompt evolution (Tier 2), immutable risk rails (Tier 3).
- Full audit trail - every decision, block reason, cost, and governance action is JSONL-logged.

2. System architecture

2.1 Data flow

External data (Alpaca, yfinance/IEX, SEC EDGAR, FRED, news sentiment, COT) feeds three agents: Skim Swarm, Unified AI, and Domain Ingest. Agents write to JSONL/JSON stores under data/. The Flask dashboard (dashboard/ai_command_center.py) aggregates state and serves the Neural Ops UI v2.

2.2 Refresh tiers

Tier	Interval	Mechanism	Panels affected
SSE	~5s	GET /api/stream/decisions	Header status, AI Mind, positions

Tier	Interval	Mechanism	Panels affected
Fast	60s + SSE	GET /api/ai/current_state	Market pulse, portfolio, beliefs
Medium	60s	Partial state patch	Domain intel, screener, LLM spend
Slow	300s	Ingest, SI, governance APIs	Ingest health, self-improvement, prompt evolution
Skim	45s	GET /api/skim/status	Skim swarm panel

Server-side cache: `build_current_state()` TTL ~25s.

3. Skim Swarm - core intraday engine

3.1 Universe (default 16 symbols)

SPY, NVDA, MSFT, GOOG, AMZN, AAPL, SOXX, NASA, BRK.B, AGIX, AVGO, LLY, V, MA, PLTR, CRWD

Each symbol is an independent agent with:

- **4 patterns:** rip_fade (short), pullback_uptrend (long), momentum_long, momentum_short
- **Per-symbol params:** entry deltas, target multiplier, disable_patterns, side pauses, causation blocks
- **Dual-timescale learning:** 10-year daily historical seeds + live 1-minute session stats
- **Phase 1 goal:** $\geq 75\%$ of enabled patterns with positive net PnL (not trade win rate)
- **Phase 2 (deferred):** Karpathy-style autoresearch after enough symbols pass Phase 1 on live paper

3.2 Recursive self-improvement loop

Every 10 exits (then every 5), per symbol:

1. Disable PnL-negative patterns
2. Pause toxic sides (pause_long / pause_short)
3. Full symbol pause if session bleeds (pause_entries)
4. Tighten entry thresholds via causation keys (pattern|side|spy_regime|sym_regime|score_bucket)
5. Boot refresh restores historical seeds; live cannot undo seeds without strong recovery evidence

4. Dashboard panels - detailed reference

Panel 0: Build stamp banner

Purpose: Deployment verification - confirms the browser loaded the expected UI bundle.

Metric	Source	Why it matters
ui_build	Env FORTRESS_AI_DASHBOARD_BU ILD	Prevents reviewing stale UI after deploy

Panel 1: Header bar

Purpose: Operational command center - instance identity, neural state, portfolio headline, operator controls.

Metric	Field	Source	Trading importance
Instance name	state.instance	FORTRESS_INSTANCE_ NAME	Multi-VM identification
Mode badge	state.dry_run	FORTRESS_AI_DRY_RU N	DRY-RUN = no live orders

Metric	Field	Source	Trading importance
Skim universe count	skim_preview.universe.length	FORTRESS_SKIM_UNIVERSE	Active basket size
Neural state	state.ui_status	_infer_ui_status()	WAITING -> OBSERVING -> THINKING -> EXECUTING
Portfolio equity	state.portfolio.equity	Alpaca get_account()	Primary capital metric
API spend today	state.today_llm_spend_usd	ai_llm_cost_ledger.jsonl	LLM cost control
Halt button	POST /api/operator/halt	operator_trading_halt.json	Emergency kill switch
Run AI cycle	POST /api/agent/run-cycle	On-demand unified agent	Manual trigger

Neural state values: EXECUTING (recent fill), THINKING (decision pending), OBSERVING (alive, idle), WAITING (off-hours/manual).

Panel 2: Skim Swarm (live)

API: GET /api/skim/status (45s poll)

Purpose: Real-time view of the always-on intraday skim basket - primary production path.

Summary tiles

Metric	Source	Computation	Trading importance
Daily realized	decisions.jsonl	Sum executed exit PnL in ET session	Today's closed P&L
Daily unrealized	Alpaca positions	Open position mark-to-market	Intraday drift
Session net	Computed	realized + unrealized	Total session P&L
Exit count	decisions.jsonl	Exit events today	Adaptation sample size
Open positions	Alpaca + state	Positions in universe	vs max_open (default 6)
HALTED	swarm_state.json	Daily stop gate (-\$200 default)	Risk circuit breaker

Per-symbol table

Column	Source	Trading importance
Side	Alpaca / agent state	Directional exposure
Price / Chg %	1m IEX/Alpaca bars	Intraday momentum
Unrealized / Realized	Alpaca + learned/*.json	Per-symbol P&L
Wins / Losses / Exits	learned/*.json	Adaptation input
Last action	state/{sym}.json	Latest agent decision

Panel 3: Why no trade?

API: trading_diagnostics in current_state (14-day lookback)

Purpose: Diagnostic - why blocked/skipped trades occurred.

Skim column key metrics: waves, entry proposed/executed/rate, HALTED, day P&L, **block_reason_counts** (pattern_disabled, pause_entries, spread_too_wide, causation_blocked, cooldown, no_edge, eod_force_flatten).

Fortress AI column: dry_run, min_confidence, executed/cycles, block_reason_counts from ai_decisions.jsonl.

Computation: `utils/trading_diagnostics.py` normalizes reasoning strings into categorical buckets.

Panel 4: AI Mind (column 1)

Purpose: Unified LLM Agent reasoning, confidence, beliefs.

Section	Source	Trading importance
Market assessment	<code>ai_decisions.jsonl</code>	LLM regime read
Chain of thought	Same	Audit trail
Confidence	0-1 score	vs min_confidence gate
Trade belief memory (P7)	<code>beliefs/beliefs.json</code>	Persistent learning cross-session
Recent decisions (5)	<code>decisions tail</code>	Quick history

Note: Unified AI does not trade skim-universe symbols (`FORTRESS_AI_SYMBOL_DENYLIST`).

Panel 5: Market pulse / Active scan / Positions (column 2)

Section	Key metrics	Source
Market pulse	SPY price/change, VIX, RSI(14)	yfinance, charts API
Active scan	Symbol chips, screener source	AI screen_market or Classic fallback
Positions	Sym, Qty, Entry, Last, P&L	Alpaca <code>get_all_positions()</code>

Panel 6: Performance / API usage / System / Domain intel / Ingest (column 3)

Section	Key metrics	Source
Performance	LLM spend chart (14d)	<code>ai_llm_cost_ledger.jsonl</code>
API usage	Today/week spend, cap bar, calls	<code>api_costs.py</code> , weekly cap env
System checklist	Kill switch, Alpaca, SSE, cap	operator halt, portfolio
Domain intel	Regime/strategy hints, concept count	<code>knowledge/intel.py</code> , ingest
Ingest health	SEC/FRED/News/COT status	<code>ingest_health.json</code>

Panel 7: Expert mode (overlay)

Trigger: Header Expert or key E | **API:** `GET /api/expert/bundle`

Tabs: Prompt note, last decision JSON, cost ledger tail, system logs, raw state JSON.

Panel 8: Self-improvement (Tier 0-1)

APIs: `/api/self_improvement/` | *Engine.* self_improvement_engine.py*

Parameter	Bounds	Effect
<code>confidence_threshold</code>	0.6-0.95	LLM entry gate
<code>decision_interval</code>	120-1800s	Loop sleep
<code>rsi_entry_threshold</code>	35-50	RSI entry hint

Flow: Propose -> Shadow -> Approve/Reject -> Monitor -> Auto-revert if win rate < 0.75. Velocity: max 1

change/7d, 3/31d.

Panel 9: Prompt evolution (Tier 2)

APIs: /api/prompt_evolution/* | Additive appendix only - cannot modify schema or risk params.

Metrics: active appendix, A/B test status, pending proposal, event log.

Panel 10: Governance (Tiers 0-3)

	Tier	Risk	Behavior
0		Low	Auto-approve if shadow criteria met
1		Medium	24h human veto window
2		Prompt	Human approval via prompt evolution
3		Immutable	Blocked - position size, stops, exposure

Panel 11: Classic vs AI comparison drawer

API: GET /api/comparison | Side-by-side Fortress AI vs Classic (trading-bot) on separate Alpaca accounts.

Metrics: equity, realized/unrealized P&L, win rate, trades/cycles today, opportunity flags, symbol overlap.

Panel 12: SPY Intraday (not mounted in live UI)

Template and APIs exist; typically disabled when skim swarm is primary.

5. Data architecture

Path	Written by	Read by
data/skim_swarm/decisions.jsonl	Skim agent	Skim panel, diagnostics
data/skim_swarm/learned/*.json	Symbol learning	Per-symbol stats
data/ai_decisions.jsonl	Unified AI	AI Mind
data/ai_llm_cost_ledger.jsonl	API costs	API usage
data/beliefs/beliefs.json	Belief manager	Trade belief memory
data/operator_trading_halt.json	Operator	Kill switch

6. Security and risk controls

Control	Mechanism
Kill switch	operator_trading_halt.json
Daily stop	FORTRESS_SKIM_DAILY_STOP_USD (-\$200)
Max open positions	Default 6
Spread filter	FORTRESS_SKIM_MAX_SPREAD_BPS (25)
EOD flatten	Force flat before close
Weekly LLM cap	FORTRESS_AI_WEEKLY_COST_CAP_USD
Symbol denylist	Unified AI blocked from skim universe

7. Known limitations

1. No billing module - LLM cost tracking only.
2. Win rate placeholder until sufficient closed trades.
3. SPY panel unwired when skim is primary.
4. Historical verify uses daily bars; live skim uses 1-minute bars.
5. Paper trading on Alpaca paper accounts.

8. Roadmap

Phase	Status	Goal
Phase 1	Active	$\geq 75\%$ winning-pattern share per symbol
Phase 2	Deferred	Autoresearch after FORTRESS_SKIM_AUTORESEARCH_MIN_WINNING_SYMBOLS gate

9. Glossary

Term	Definition
Pattern	rip_fade, pullback_uptrend, momentum_long, momentum_short
Winning pattern	Positive net PnL after min samples - not trade win rate
Wave	One skim loop iteration across universe
Causation key	Context bucket blocking repeated losing entries
Belief (P7)	Persistent learned fact from closed trades
Neural state	WAITING / OBSERVING / THINKING / EXECUTING